higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 2

13 NOVEMBER 2019

This marking guideline consists of 5 pages.

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SECTION A

QUESTION 1

1.1 False
1.2 False
1.3 True
1.4 True
1.5 False

(5 × 1) **[5]**

QUESTION 2

2.1 C
2.2 E
2.3 A
2.4 D
2.5 B

(5 × 1) **[5]**

QUESTION 3

3.1 Sapwood
3.2 Hot loop
3.3 Humidifying
3.4 Wet end
3.5 Screen printing

(5 × 1) **[5]**

TOTAL SECTION A: 15

QUESTION 4

4.1 Papyrus (1)

4.2 4.2.1
- Drying paper, board and pulp
- Heating chips in TMP refiners; heating chemical pulp digesters
- Generating electrical power (Any 2 × 1) (2)

4.2.2
- By burning waste liquor
- By burning bark and wood waste (2)

4.3 De-inking is the industrial process of removing printing ink✓ from fibres of recycled paper to make de-inked pulp.✓ The de-inking process has the ability to detach ink from the fibres.✓ This is achieved by a combination of mechanical✓ action and chemical means.✓ (5)

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4.4
- Seed hairs
- Bast fibres
- Leaf fibres
- Wood fibres
- Grass fibres

Any (4 × 1) (4)

4.5
- Compression wood
- Tension wood

(2)

4.6 Micro-organisms like fungi, bacteria, yeast and decay (1)

[17]

QUESTION 5

5.1 5.1.1 It is the removal of partially bonded material from the paper surface during printing.

5.1.2 It is the slow removal of loose fibre from the paper surface, which accumulates on the printing blankets and inking units.

5.1.3 It is the complete removal of the surface layers, or delamination of inner layers within the sheet structure.

(2 × 3) (6)

5.2
- Fibre furnish
- The level of filler addition
- The type of Fourdrinier wire and press felts used
- The degree of pressing and calendering
- The surface smoothness of press and dryer rolls
- Finishing operations like sizing and coating
- The compressibility of the paper
- Moisture content
- Degree of refining

(Any 4 × 1) (4)

5.3
- The storage area must be open and well drained.
- The ground should be surfaced with gravel, crushed rocks, blacktop or concrete.
- The storage area must be kept free of weeds and debris that restrict air movement along the surface of the ground, create a place where insects and fungi live, or create a fire hazard when dry.

(3)

5.4
- The shape or surface of the wood.
- The amount of work needed to break the bond between the bark and the wood.
- The amount of work needed to break the bond between the bark fibres themselves.

(3)

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5.5
- Rocks
- Tramp metals
- Plastics
- Sand and grit (4)

[20]

QUESTION 6

6.1

A	White liquor feed tank
B	Rupture or safety disc
C	Digester cap
D	Vapour relief
E	Heat exchanger
F	Circulation pump
G	Blow valve
H	Blow line
I	Digester shell
J	Black liquor feed tank (10)

6.2 It is a chemical substance✓ used for chemical pulping.✓ (2)

6.3 Raw limestone ($CaCO_3$) is crushed✓ and converted into slurry (lime mud) for pumping and storage purposes.✓ The slurry is first dewatered by means of mechanical and thermal methods.✓ The dry limestone is then fed into a kiln where it is calcined.✓ The hot unslaked lime is cooled,✓ screened, crushed✓ and stored.✓ (7)

6.4
- Mills were small and the cost of the recovery unit overshadowed the value of the cooking liquor.
- The environmental implications of dumping these chemicals were not appreciated.
- Poor process and equipment design made it difficult to recover these chemicals. (3)

6.5 It makes it possible to use the latent heat of vaporisation in the inlet live steam several times over, and reduces overall steam requirement. (2)

[24]

QUESTION 7

7.1 The approach flow system is used to dilute the thick stock,✓ clean and screen the stock, remove air from the stock and supply a continuous✓ and uniform stock flow to the head box.✓ (3)

7.2
- Remove water from sheet
- Consolidate the sheet of paper
- Improve the sheet surface quality (3)

7.3
7.3.1 Dry end section
7.3.2 Press section
7.3.3 Forming section
7.3.4 Dry end section
(4 × 1) (4)

7.4 This is the part of the paper machine where the paper is dried, surface sized, calendered and reeled. (2)

7.5
- Aluminium foil with paper
- Aluminium foil with paper board
- Paper with paper
- Paper with paper board
- Paper to scrim to paper
- Paper to scrim to aluminium foil (Any 5 × 1) (5)

7.6 Coating and treating the surface of paper or board improves quality, smoothness and gloss for improved appearance and printability. (2)

7.7
- No retained solvent
- No odour or taste
- Fast curing
- No tunnelling, which speeds up further conversion
- Direct coating on webs that are sensitive to solvents
- Low energy demand in drying by using a low volume of heated air that can be recirculated (Any 5 × 1) (5)

[24]

TOTAL SECTION B: 85
GRAND TOTAL: 100

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